

Muhammad Mursaleen

AI & ML Engineer

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Professional Summary

AI & ML Engineer with 2+ years of experience building production-grade LLM applications, RAG systems, and agent workflows. Delivered FastAPI-based AI platforms serving healthcare, finance, and Islamic knowledge domains using LangGraph, Qdrant, AWS, and Docker. Experienced in fine-tuning, retrieval optimization, and scalable AI backend development.

Experience

AI/ML Engineer

DiveDeepAI

March 2025 – Present

Islamabad, Pakistan

- Engineered Generative AI solutions, developing both custom and LangGraph driven RAG pipelines, and fine-tuning LLMs.
- Architected robust FastAPI backends; managed deployments on AWS EC2 and DigitalOcean.
- Led a 3-person team to deliver sprint goals, managing system architecture and mentored junior developers.

Jr AI Engineer

Softoo

August 2024 – February 2025

Islamabad, Pakistan

- Collaborated with the Solution Architect to rapidly prototype and iterate Generative AI Proof-of-Concepts (POCs).
- Built scalable RAG-based applications utilizing LangChain, LangGraph, LLMs, and vector databases.

Projects

Bridge Vet: Veterinary Clinic Platform

- Managed a development team to deliver an end-to-end clinic management system; built a scalable FastAPI, Redis, and Celery backend for asynchronous AI SOAP note generation, medication extraction, and appointment follow-ups.
- Developed a custom RAG-based clinical assistant for dosage calculations, evaluated retrieval accuracy via Langfuse, and implemented OCR pipelines to automate inventory tracking. Maintained continuous deployments on AWS EC2.

Tech Stack: Python, FastAPI, Redis, Celery, Google Gemini, Vector Databases, Langfuse, OCR, AsyncIO, Docker, AWS EC2

eQRA-AI: Islamic AI Assistant Mobile App

- Built the backend for an Islamic AI Assistant, deploying dual RAG chatbots (Phase 1) dedicated to precise Fatwa and Duas retrieval. Scaled deployments on DigitalOcean App platform and Redis/Celery over droplets and Grafana Loki for log aggregation.
- Developed an agentic MVP (Phase 2) using a LangGraph multi-tool agent to efficiently route user intent between live web queries and vectorized Qdrant databases. Integrated Mem0 for long-term memory and built web scraping pipelines to ingest extensive Islamic literature.

Tech Stack: Python, FastAPI, LangGraph, Qdrant, Mem0, Redis, Celery, Grafana Loki, Web Scraping, AWS, Docker, DigitalOcean

Yani: AI Assistant for Chartered Accountants

- Built a RAG-based chatbot to assist Chartered Accountants, alongside a distinct document compliance workflow that evaluates text chunks directly against regulatory frameworks to map clauses and flag non-compliance.
- Implemented an AI-powered database chat utilizing Vanna.ai trained on complex schemas, achieving 92% accuracy in natural language to SQL conversion.
- Developed a multi-agent system using Autogen to perform automated Exploratory Data Analysis (EDA) and dynamically generate visual insights from the extracted SQL data.

Tech Stack: Python, Vanna.ai, Autogen, RAG, LLMs, Pandas, Matplotlib, Streamlit

Fine-Tuning Whisper for Quranic Arabic Transcription

- Optimized a custom fine-tuning pipeline for the Whisper model utilizing 10,000+ Quranic Tarteel audio samples to improve specialized Arabic speech-to-text transcription and phonetic accuracy.
- Achieved a 90% improvement in complex Harakat (diacritic) recognition and reduced overall Word Error Rate (WER) by 30–40% compared to foundational baseline models.

Tech Stack: Python, PyTorch, Hugging Face, Whisper, Audio Processing

Education

The Islamia University of Bahawalpur

Bachelor of Science in Software Engineering; CGPA: 3.70/4.00

Punjab, Pakistan

Oct. 2020 – June 2024

Skills

Languages: Python, SQL

AI/ML & Frameworks: PyTorch, Hugging Face, OpenAI, LangChain, LangGraph, Autogen, RAG, Mem0

Evaluation & Observability: Langfuse, Grafana Loki, LLM Benchmarking, Retrieval Evaluation, Hallucination Reduction

Databases: PostgreSQL, Vector DBs (Qdrant, FAISS, PgVector), Redis, MongoDB

Backend & Infrastructure: FastAPI, Celery, AWS (EC2, S3), DigitalOcean, Docker, Linux, CI/CD

Tools: Git, GitHub, GitLab, Web Scraping, Postman, Claude Code, Cursor, Antigravity.